Financial Functions Analysis Using Excel & Power BI

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Internship Project Report

Abstract

This report presents a detailed financial analysis project completed using Microsoft Excel and Microsoft Power BI. The project evaluates an investment using financial functions such as Net Present Value (NPV), Return on Investment (ROI), Payback Period, Total Investment, Total Returns and Net Profit. The dashboard was designed to provide decision-makers with an interactive summary of investment performance.

Introduction

Financial analysis is one of the most important activities in business decision making. Organizations evaluate investments before allocating capital. Excel provides powerful financial formulas while Power BI helps transform results into interactive dashboards. In this project, yearly cash flow data was analyzed to understand the profitability of an investment.

Objectives

• Analyze the financial dataset.
• Calculate NPV, ROI, Payback Period, Total Investment, Total Returns and Net Profit.
• Build an interactive Power BI dashboard.
• Generate business insights and recommendations.

Dataset Description

The dataset contains Year 0 as the initial investment of -5000 and yearly positive cash inflows for Years 1–5. The investment is evaluated using a 10% discount rate.

Tools Used

Microsoft Excel was used for calculations and Microsoft Power BI was used for dashboard creation. Excel formulas were used to calculate NPV and other financial metrics while Power BI visualized KPI cards, charts and summaries.

Methodology

The workflow included data loading, data validation, financial calculations, importing the cleaned data into Power BI, creating KPI cards, designing charts and interpreting the results.

Financial Calculations

NPV = 3251.80
Total Investment = 5000
Total Returns = 10869
Net Profit = 5869
ROI = 117.38%
Payback Period = 2.30 Years

The positive NPV indicates that the investment creates value. ROI above 100% indicates the investment generated returns greater than the initial investment. The payback period shows the investment is recovered in approximately 2.3 years.

Dashboard Description

The dashboard includes KPI cards for all financial metrics, a year-wise cash flow chart, profitability analysis, and an investment summary table. These visuals make the financial performance easy to interpret.

Results and Analysis

Cash flows become positive after the initial investment. The project demonstrates strong financial performance because total returns exceed the investment and the calculated NPV is positive. The dashboard clearly highlights yearly trends and overall profitability.

Key Insights

1. Positive NPV indicates a profitable investment.
2. ROI of 117.38% reflects excellent returns.
3. Total returns exceeded total investment by 5869.
4. Investment recovery occurs within approximately 2.3 years.
5. The dashboard enables quick financial decision-making.

Recommendations

Investments with positive NPV and high ROI should be considered for future funding. Businesses should continuously monitor cash flows and compare actual returns with projected values. Interactive dashboards should be updated regularly.

Challenges

Challenges included preparing data, selecting appropriate visuals, validating formulas and presenting financial metrics in an understandable format.

Learning Outcomes

This project improved skills in Excel financial functions, Power BI dashboard development, financial interpretation, KPI reporting and business intelligence.

Future Scope

Future versions may include IRR, sensitivity analysis, forecasting, scenario analysis, dynamic slicers and real-time financial data.

Conclusion

The project successfully demonstrated the use of Excel and Power BI for financial analysis. All required financial functions were calculated accurately and presented through an interactive dashboard. The investment is financially attractive based on the positive NPV, strong ROI and reasonable payback period.

Detailed Discussion

Financial analysis supports strategic planning by helping organizations estimate the value created by investments. Each financial metric provides a different perspective. NPV measures value after considering the time value of money. ROI evaluates efficiency, while the payback period focuses on liquidity and investment recovery. Power BI enhances decision-making by combining these metrics into a single interactive dashboard. In real business environments, analysts also compare multiple investment alternatives, evaluate risks, monitor trends over time, and communicate findings using dashboards and reports. Financial analysis supports strategic planning by helping organizations estimate the value created by investments. Each financial metric provides a different perspective. NPV measures value after considering the time value of money. ROI evaluates efficiency, while the payback period focuses on liquidity and investment recovery. 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